CSCE 4813 – Midterm Exam
Spring 2020

Student Name: Solutions

Student UAID: _____

$\frac{60}{60}$

Instructions:

- This is a one hour exam.
- Students <u>are</u> allowed one 8.5 by 11 page of study notes.
- Electronics (calculators, cell phones, computers, etc.) are <u>not</u> allowed.

Reminder:

This class is governed by the code of ethics of the College of Engineering and Department of Computer Science and Computer Engineering. Any instances of cheating will result in a grade of zero on the exam and referral to the ethics board.

Geometric Transformations (15 points)

[5 points] Consider the problem of rotating all of the polygons in the scene by an angle of theta around the Z axis. Which of the following formulas would transform the (x,y,z) coordinates of these polygons into (x',y',z')?

A) x' = x*sin(theta); y' = y*cos(theta); z' = z;
B) x' = x*cos(theta); y' = y*sin(theta); z' = z;
C) x' = x*cos(theta) – y*sin(theta); y' = x*sin(theta) + y*cos(theta); z' = z;
D) x' = x*sin(theta) – y*cos(theta); y' = x*cos(theta) + y*sin(theta); z' = z;
E) None of the above.

[5 points] Consider the problem of transforming the polygons in the scene by a translation of (Tx,Ty,Tz) <u>followed</u> by a scaling by (Sx,Sy,Sz). Which of the following formulas would transform the (x,y,z) coordinates of these polygons into (x',y',z')?

A) x' = Tx + x*Sx; y' = Ty + y*Sy; z' = Tz + z*Sz;
B) x' = Sx + x*Tx; y' = Sy + y*Ty; z' = Sz + z*Tz;
C) x' = Tx + (1+x)*Sx; y' = Ty + (1+y)*Sy; z' = Tz + (1+z)*Sz;
D) x' = Tx*Sx + x*Sx; y' = Ty*Sy + y*Sy; z' = Tz*Sz + z*Sz;
E) None of the above.

[5 points] In this question, T(tx,ty,tz) will denote a 4x4 translation matrix, S(sx,sy,sz) will denote a 4x4 scaling matrix, and Rz(theta) will denoted a 4x4 Z rotation matrix. If the center of an object is located at (Cx, Cy, Cz), what sequence of matrix operations will rotate this object <u>around its center</u> by theta degrees around the Z axis?

A) (x', y', z') = T(Cx, Cy, Cz) * Rz(theta) * T(Cx, Cy, Cz) * (x, y, z)
B) (x', y', z') = T(Cx, Cy, Cz) * Rz(theta) * (x, y, z)
C) (x', y', z') = T(Cx, Cy, Cz) * Rz(theta) * T(-Cx, -Cy, -Cz) * (x, y, z)
D) (x', y', z') = T(-Cx, -Cy, -Cz) * Rz(theta) * T(Cx, Cy, Cz) * (x, y, z)
E) None of the above.

Perspective and Parallel Projection (15 points)

[5 points] Consider the problem of performing perspective projection with the center of projection is at (0,0,0) and the projection plane is z=d, where d is positive. What is the formula for projecting the point (x,y,z) onto the projection plane at (x',y',z')?

A) x'=x*z/d; y'=y*z/d; z'=d;
B) x'=x*d/z; y'=y*d/z; z'=d;
C) x'=x*(z+d); y'=y*(z+d); z'=d;
D) x'=x*(z-d); y'=y*(z-d); z'=0;
E) None of the above.

[5 points] Consider the problem of performing perspective projection with the center of projection at (0,0,0) and the projection plane at z=d, where d is positive. What will happen if we translate an object at location (x,y,z) to (x+Tx, y+Ty, z+Tz)?

A) It will appear larger if Tx > 0
B) It will appear larger if Ty > 0
C) It will appear smaller if Tz < 0
D) It will appear to be the same size.
E) None of the above.

[5 points] Consider the problem of performing parallel projection in OpenGL. If we set the window size with glutInitWindowSize(500, 500); and specify the viewing window with glOrtho(-1.0, 1.0, -1.0, 1.0, -1.0, 1.0); where will the 3D point (0.5, 0.5, 0.5) appear in the display window?

A) x' = 0.5 y' = 0.5
B) x' = 500 * 0.5 = 250 y' = 500 * 0.5 = 250
C) x' = 500 * (0.5 +1) = 750 y' = 500 * (0.5 + 1) = 750
D) x' = 500 * (0.5 +1) / 2 = 375 y' = 500 - 500 * (0.5 + 1) / 2 = 125
E) None of the above.

Geometric Model Construction (15 points)

[5 points] Consider the problem of displaying a bar chart in a 500x500 display window. If there are 10 evenly bars in your chart, and you want to display data values in the range [min_value .. max_value], how should you calculate the height of each bar?

A) Height = 500 * (value – min_value)
B) Height = 500 * (max_value – min_value) / (value – min_value)
C) Height = 500 * (value – min_value) / (max_value – min_value)
D) Height = (500 – value) / (max_value – min_value)
E) None of the above.

[5 points] Consider the problem of modeling a sphere using a 100x100 polygon mesh. If the (x,y) points on the "side view" of the sphere are given by x=cos(theta) and y=sin(theta) and z=0 for theta in the range [-PI/2 .. PI/2], then how can we calculate (x,y,z) for each of the 100x100 points on the surface.

A) Calculate 100 points on "side view" and rotate in 100 steps around X axis.
B) Calculate 100 points on "side view" and rotate in 100 steps around Y axis.
C) Calculate 100 points on "side view" and rotate in 100 steps around Z axis.
D) Calculate 50 points on "side view" and rotate in 200 steps around Z axis.
E) None of the above.

[5 points] Consider the problem of calculating the surface normals for an object represented by a polygon mesh. If P(i,j) represents the (x,y,z) coordinates of a point at location (i,j) on the mesh, we can define two surface tangents T1(i,j) = P(i+1,j) – P(i,j) and T2(i,j) = P(i,j+1) – P(i,j). Which formula below can be used to calculate the <u>surface normal</u> at each point on the surface?

A) N(i,j) = T1(i,j) + T2(i,j) where + denotes vector addition
B) N(i,j) = T1(i,j) * T2(i,j) where * denotes dot product
C) N(i,j) = T1(i,j) x T2(i,j) where x denotes cross product
D) N(i,j) = T1(i,j) * P(i,j) + T2(i,j) * P(i,j) where * denotes dot product
E) None of the above.

Computer Shading (15 points)

[5 points] Consider the problem of implementing Phong shading where:

P = (Px,Py,Pz) = the point on the surface we are shading.
N = (Nx,Ny,Nz) = the surface normal at this point.
V = (Vx,Vy,Vz) = the location of the viewer.
R = (Rx,Ry,Rz) = the ideal reflection direction.
L = (Lx,Ly,Lz) = the direction of the light source.
p = a power chosen by the user

What is the formula needed to calculate the <u>diffuse</u> component?

A) $P \cdot N = (Px*Nx + Py*Ny + Pz*Nz)$
B) $(R \cdot V)^p = (Rx*Vx + Ry*Vy + Rz*Vz)^p$
C) $(L \cdot N)^p = (Lx*Nx + Ly*Ny + Lz*Nz)^p$
D) $N \cdot L = (Nx*Lx + Ny*Ly + Nz*Lz)$
E) None of the above.

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What is the formula needed to calculate the <u>specular</u> component?

A) $P \cdot N = (Px*Nx + Py*Ny + Pz*Nz)$
B) $(R \cdot V)^p = (Rx*Vx + Ry*Vy + Rz*Vz)^p$
C) $(L \cdot N)^p = (Lx*Nx + Ly*Ny + Lz*Nz)^p$
D) $N \cdot L = (Nx*Lx + Ny*Ly + Nz*Lz)$
E) None of the above.

[5 points] Consider the problem of implementing Phong shading with N light sources. Assume that you know the locations and colors of each of these N lights, and you are able to use the Phong shading formulas above to calculate the (r,g,b) color for point P using each of these light sources separately. Which of the following methods would produce the most realistic output color for point P?

A) output color = the (r,g,b) value from the closest light source to point P
B) output color = average of all N (r,g,b) values
C) output color = inverse distance weighted sum of all N (r,g,b) values
D) output color = sum of all N (r,g,b) values
E) None of the above.